

TASKS AND ACCOMPLISHMENTS

Albert (Leader)

Tasks Assigned:

- Set up the project repository on GitHub and established initial file structure.
- Planned project timeline with milestones.
- Researched user authentication methods for secure file access.

Accomplishments:

- Successfully set up a shared GitHub repository and established version control guidelines.
- Conducted a team meeting to align goals for the next sprint.
- Helped resolve an early bug related to file reading/writing during encryption.

Kaleb

Tasks Assigned:

- Researched AES and RSA encryption techniques and documented key points.
- Worked on integrating OpenSSL libraries for encryption.
- Started implementing AES encryption function in Python.

Accomplishments:

- Successfully implemented a basic AES encryption function.
- Wrote a small test script to generate and store RSA key pairs.
- Identified potential performance issues with encryption speed.

Jorge

Tasks Assigned:

- Created a simple CLI prototype for encrypting and decrypting files.
- Assisted in debugging encryption issues.

- Planned database schema for storing encrypted file metadata

Accomplishments:

- Developed an early CLI prototype for testing encryption functions.
- Helped fix a bug related to incorrect file handling during encryption.
- Drafted an initial structure for the database schema.

Max**Tasks Assigned:**

- Began designing the UI layout for the application.
- Researched libraries and frameworks for building the UI.
- Worked on structuring user input handling for encryption/decryption.

Accomplishments:

- Designed a rough UI wireframe for the application.
- Implemented basic user input handling for selecting files to encrypt/decrypt.
- Provided feedback on how encryption functionality should interact with the user interface.

USER STORIES AND ACCEPTANCE TESTS

User Story 1: Encrypt a File

As a user I want to encrypt a file using AES encryption so that I can securely store or share sensitive information.

Acceptance Test: Given that the user has selected a file. When they click the “Encrypt” button, then the system should encrypt the file using AES and save it with a .enc extension.

User Story 2: Decrypt a File

As a user I want to decrypt a previously encrypted file so that I can access my original data when needed.

Acceptance Test: Given that the user has an encrypted file and enters the correct password. When they click the “Decrypt” button, then the system should decrypt the file and restore it to its original format

User Story 3: Secure Key Management

As a user I want the encryption system to generate and securely store encryption keys so that I don't lose access to my encrypted files.

Acceptance Test: Given that the user is encrypting a file for the first time. When the system generates an encryption key, then the key should be securely stored or exported for future decryption.

User Story 4: Password Protection

As a user I want to use a password for encrypting and decrypting files so that only authorized users can access them.

Acceptance Test: Given that the user enters a password before encryption. When they attempt to decrypt the file, then the system should require the correct password before proceeding

User Story 5: File Integrity Verification

As a user I want the encryption system to verify file integrity after decryption so that I can be sure my data hasn't been tampered with.

Acceptance Test: Given that a user decrypts a file. When the decryption process is complete , then the system should compare the file's hash before encryption and after decryption to confirm integrity.

User Story 6: RSA Key Pair for Secure Sharing

As a user I want to encrypt files using an RSA public key, so that only the owner of the corresponding private key can decrypt them.

Acceptance Test: Given that a user selects RSA encryption. When they encrypt a file using a recipient's public key, then only the recipient with the correct private key should be able to decrypt the file

User Story 7: User-Friendly Interface

As a user I want an intuitive UI with clear encryption and decryption options, so that I can easily use the application without technical knowledge.

Acceptance Test: Given that a new user opens the application. When they navigate through the interface, then they should see clearly labeled encryption and decryption buttons, input fields, and instructions.